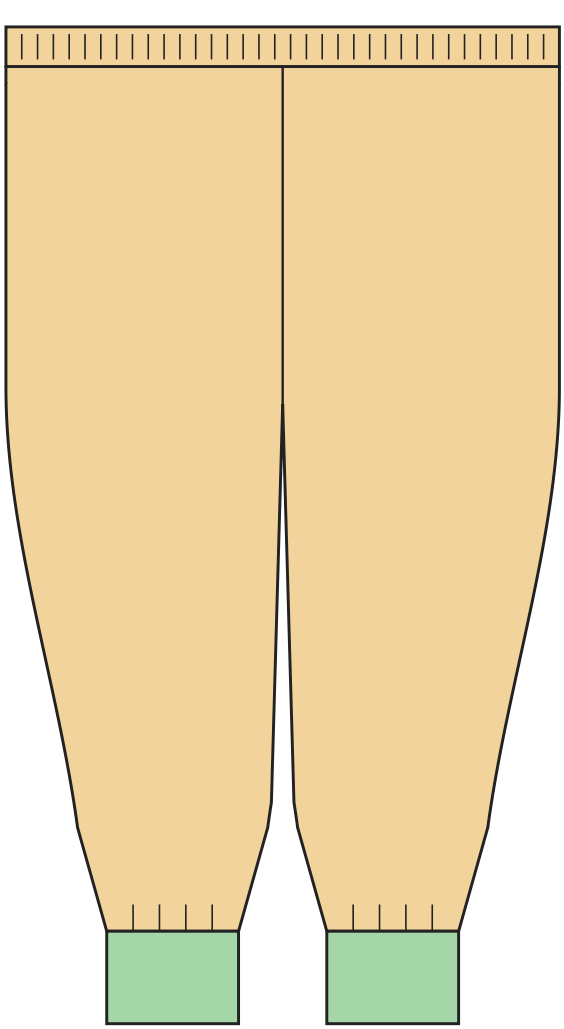
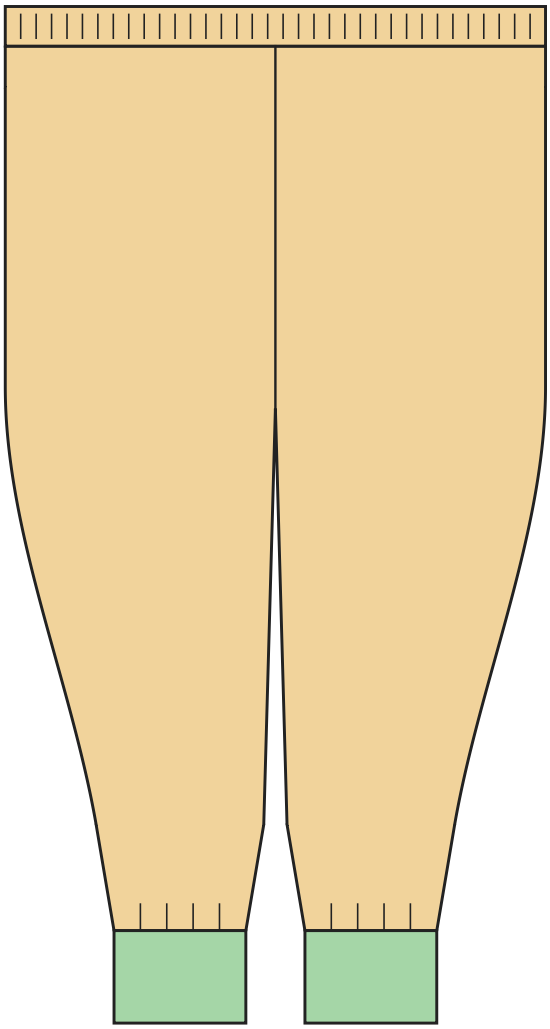
Harem pants

size 122 · harem (wide, dropped crotch) fit, Sweat / French Terry (25 % stretch)

4 parts · 20 steps · medium · about 3 h



Front view



Back view

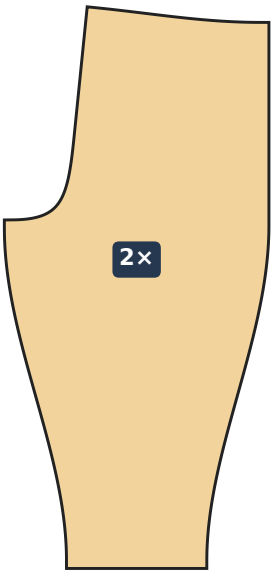
Sewn with: Sewing machine

Sewing machine

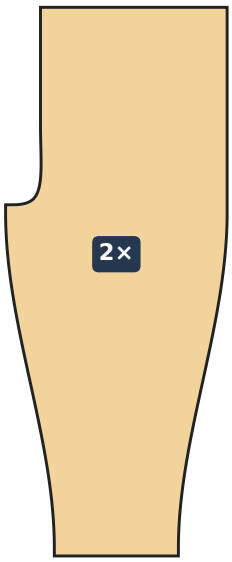
Iron

Read the steps once before you start. Every picture is drawn from your pattern pieces. Sew a test garment from cheap fabric first and adjust before you cut the good fabric.

Parts



B · Back leg



A · Front leg



D · Leg cuff



C · Waistband (elastic) (cut on fold)

Seam allowance (included): 1.0 cm

Material

Main fabric
0.90 m at 150 cm width

Rib knit
0.20 m at 70 cm width

Elastic
Elastic 20 mm wide, length 50 cm (+2 cm overlap for joining)

Sewing machine
narrow zigzag 1 × 2.5 mm (or overlocker); wide zigzag 4 × 2.5 mm; twin needle 4 mm, stitch 3 mm

Iron
steam

Polyester sewing thread
In the colour of the fabric.

Pins or clips
Pin across the seam line.

Chalk or a fabric pen
For notches, labels and placement lines.

Colours in this booklet

Main fabric
right side · wrong side

Rib knit
right side · wrong side

Sew now (machine)

Sew by hand

Already sewn

Fold / press line

Marking (chalk)

Pin

Notch: match them

Right/wrong side of the fabric = the side of the fabric. Left/right (as worn) = seen from the child wearing it: the left front leg is on the right when you look at the child.

Basics: Sewing machine

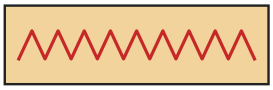
Steps are written for a sewing machine with zigzag. Have an overlocker or a coverstitch machine? Tell the generator which machines you own and the steps use them.

Basics: the sewing machine



Straight stitch

Seams in woven fabric, 2.5 mm; backstitch at the start and the end.



Zigzag

Knit seams (narrow) and finishing the edges (wide).

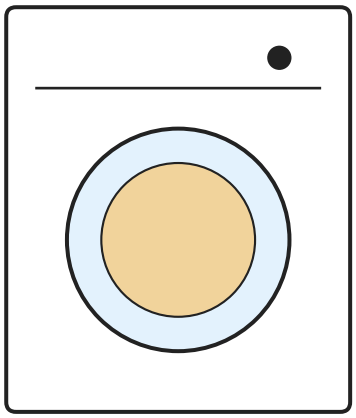


Topstitching

Right side up, with the same distance to the edge all the way; use the edge of the presser foot as a guide.

1 Pre-wash the fabric

1. Wash and dry it the way you will wash the garment later, so it shrinks now, not after sewing.



30–40°

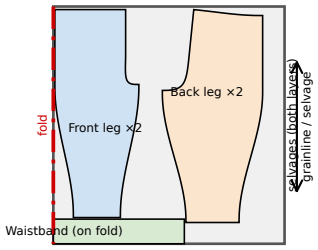
2 Cut out all pieces

Scissors

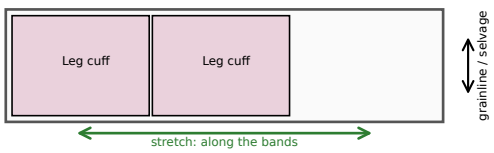
1. Follow the cutting layout
2. Grainline arrows parallel to the selvage.
3. Rib knit: the ribs run along the grainline, so every band stretches along its length (across the fabric).

Tip: Cut both layers at once with long strokes; keep the fabric flat on the table.

Main fabric · 150 cm · folded (double layer)



Rib knit · 70 cm · single layer

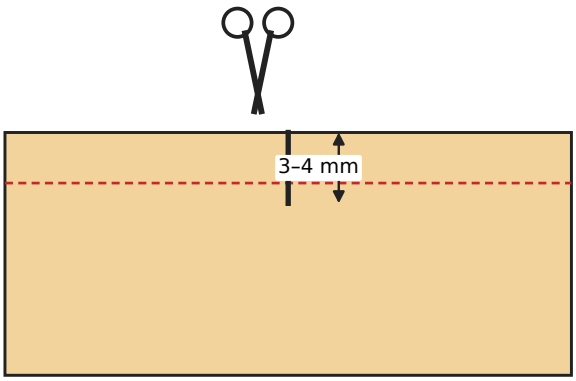


3 Snip the notches

Chalk

1. Cut 3–4 mm into the seam allowance at every notch.

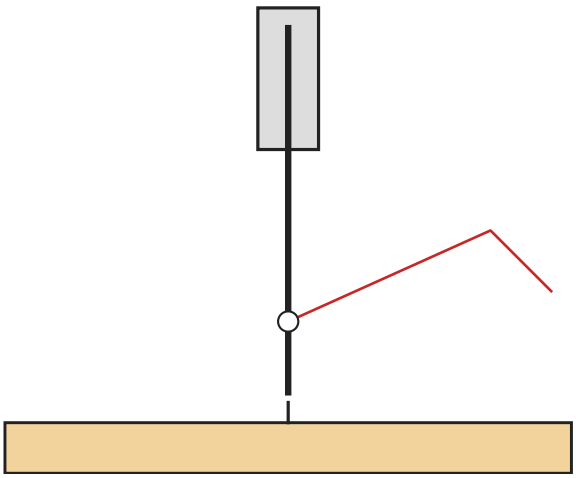
⚠ Important: Never deeper than half the seam allowance.



4 Set up the machine

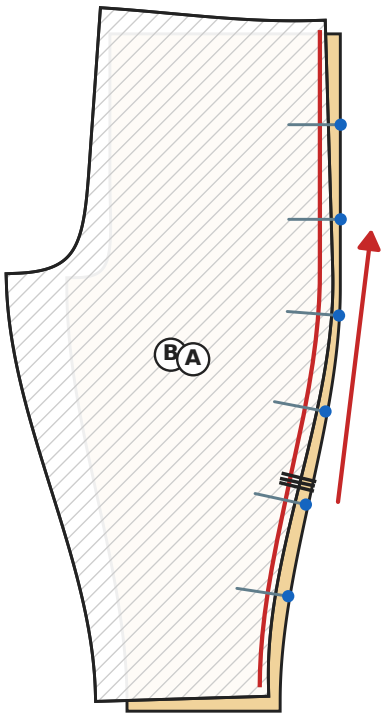
1. Sewing machine: jersey/stretch needle 75–80, polyester thread.
2. Sew a test seam on two layers of a scrap.

Tip: See the basics page at the front of the booklet.



5 Sew the side seams

Sewing machine · Pins



1. Lay front and back of the same leg right sides together, side edges level: single notches = front, double notches = back.
2. Pin, then sew with 10 mm seam allowance from the top edge to the hem.

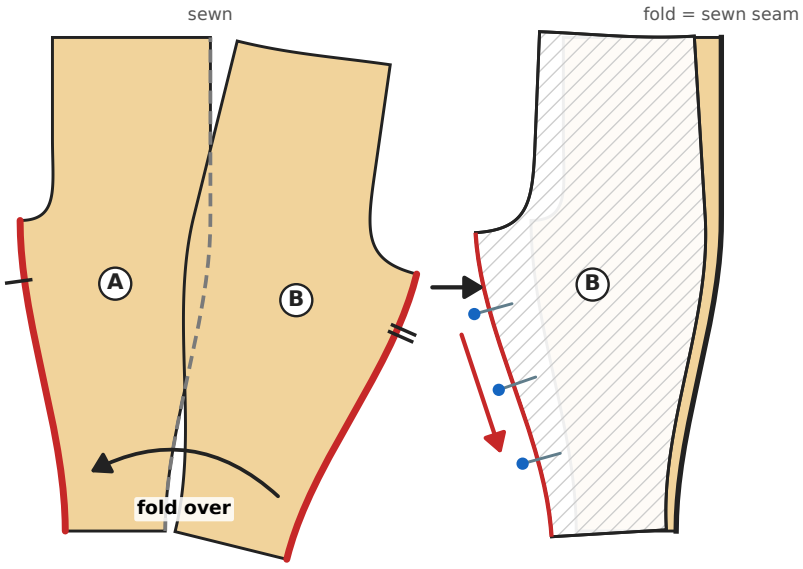
Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)

⚠ Important: Always the front and back of the SAME leg (FL + BL, FR + BR).

2×

6 Sew the inseams

Sewing machine · Pins



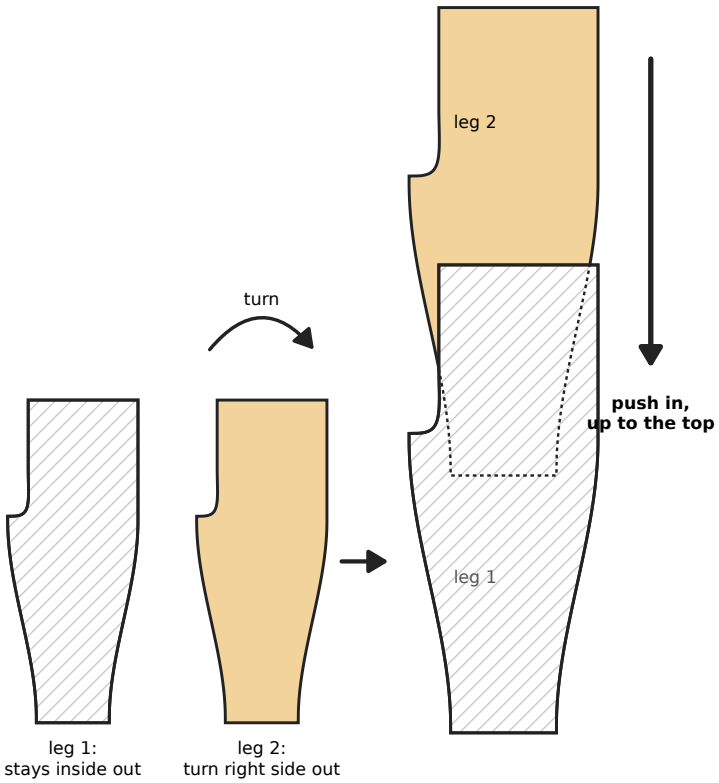
1. Open the leg out at the side seam and fold it right sides together, inseam edges level.
2. Pin the notches first, then the ends.
3. Sew with 10 mm seam allowance from the crotch point to the hem.

Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)

⚠ Important: Start exactly at the crotch point: the four seams meet there later.

2×

7 Put one leg inside the other

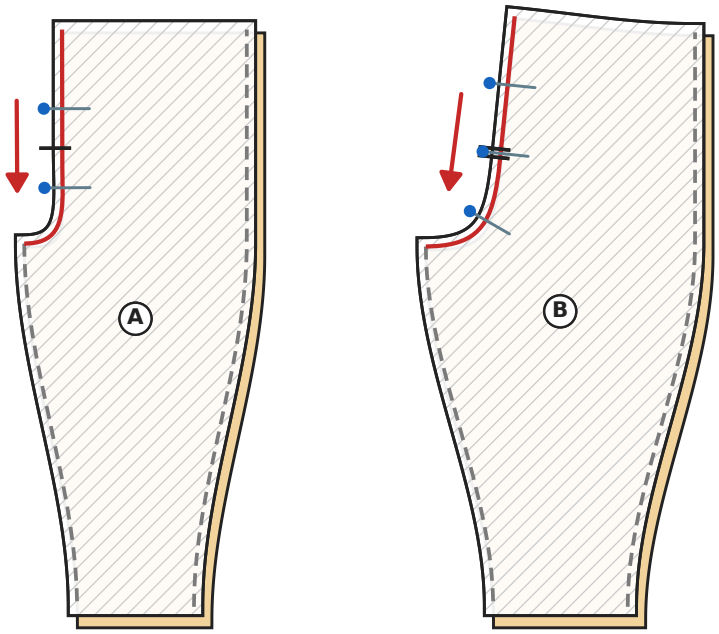


1. Turn leg 2 right side out.
2. Leg 1 stays wrong side out.
3. Push leg 2 into leg 1 until the top edges are even
4. The hems point the same way.

Tip: Now the legs lie right sides together and the crotch curves lie exactly on top of each other.

8 Sew the crotch seam





1. Pin the crotch curves together, starting at the inseams: they lie exactly on top of each other.
2. Sew with 10 mm seam allowance in one go from the front top edge to the back top edge.

Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)

⚠ Important: Sew the crotch curve a second time for strength.

Tip: Press the inseam allowances in opposite directions: the crossing lies flat. Pull the inner leg out afterwards.

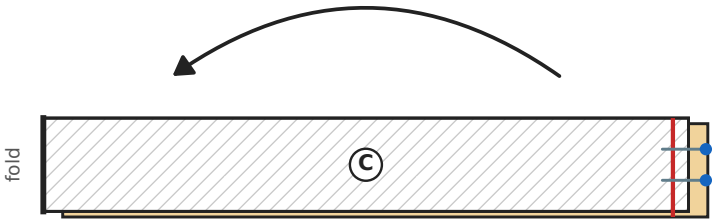
9 Close the waistband into a ring

Sewing machine

Pins

1. Fold the waistband crosswise right sides together and sew the short ends (centre back) with 10 mm seam allowance
2. Leave 3 cm open for the elastic.

Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)

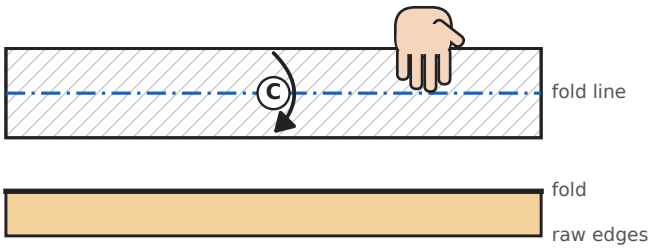


10 Fold the waistband in half

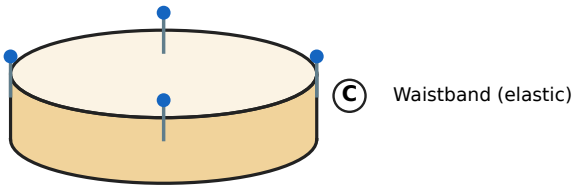
Iron

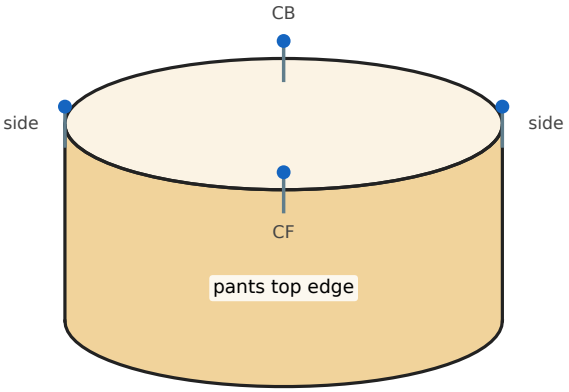
1. Fold lengthwise wrong sides together along the fold line and press.

Iron: steam



11 Mark quarters



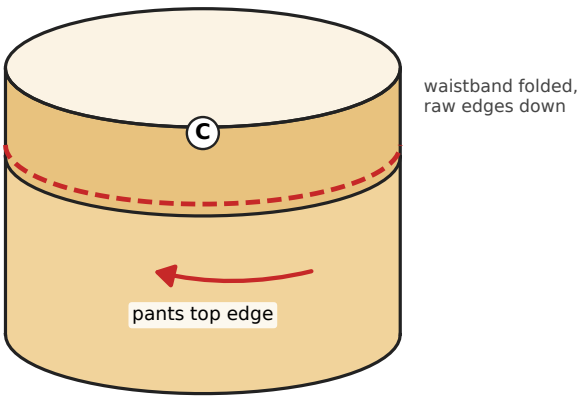


1. Mark centre front, centre back and both sides with pins, on the waistband and on the pants top edge.

12 Sew the waistband on


Sewing machine


Pins



1. Pants right side out: slide the waistband over them, raw edges level, the opening at the back on the inside.
2. Match the quarter pins.
3. Sew all round with 10 mm seam allowance.

Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)

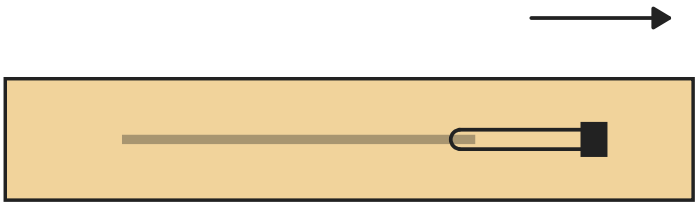
13 Thread the elastic

Sewing machine

Safety pin

1. With a safety pin through the opening
2. Overlap the ends by 2 cm and sew them together.

Sewing machine: wide zigzag 4 × 2.5 mm



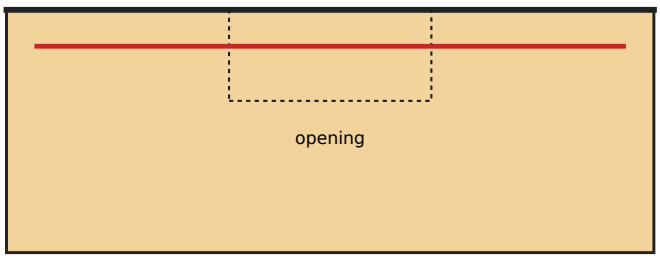
elastic, with a safety pin

14 Close the opening

Sewing machine

1. Spread the gathers evenly, close the opening and stitch in the ditch at the side seams so the elastic cannot twist.

Sewing machine: twin needle 4 mm, stitch 3 mm



stitch close to the fold, over the opening

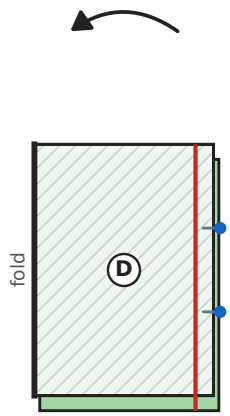
15 Close the cuffs into rings

Sewing machine

Pins

1. Fold each cuff crosswise right sides together, short edges level.
2. Sew with 10 mm seam allowance
3. Finger-press the allowances open.

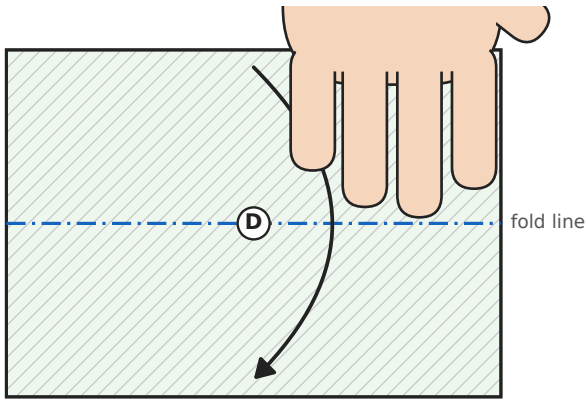
Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)



2×

16 Fold the cuffs in half

Iron



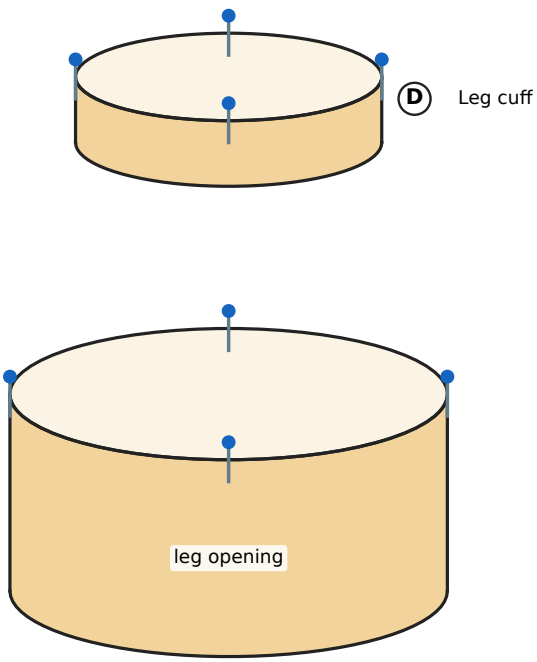
1. Fold the ring lengthwise wrong sides together, raw edges level.
2. The seam allowance is now inside
3. The right side shows all round.

Iron: steam



2×

17 Mark quarters

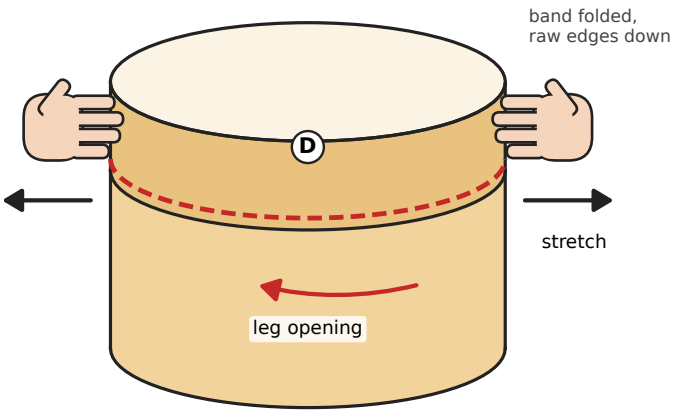


1. Mark quarters with pins on the cuff and on the leg opening
2. The cuff seam is one of them.

2×

18 Sew the cuffs on

Sewing machine · Pins



1. Turn the leg right side out.
2. Slide the cuff over the leg end, raw edges level, cuff seam on the inseam.
3. Match the quarter pins.
4. Sew with 10 mm seam allowance, cuff on top, stretching only the cuff.

Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)

Tip: One hand in front of the presser foot, one behind it: both stretch evenly.

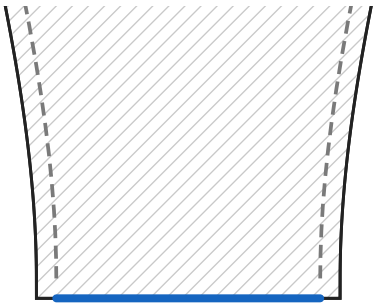
2×

19 Press the allowance up

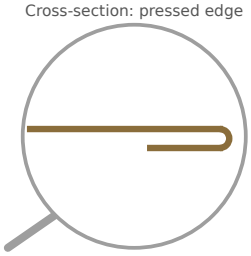
Iron

1. Press the seam allowance up towards the pants.

Iron: steam



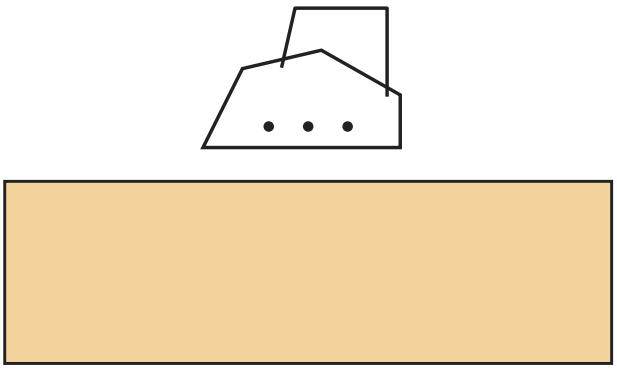
Cross-section: pressed edge



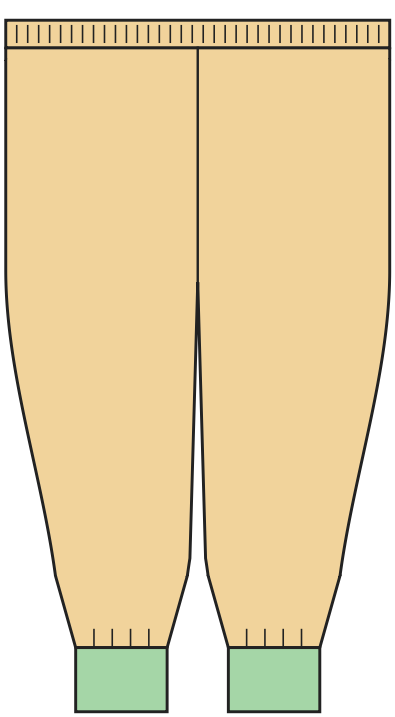
2×

20 Final pressing

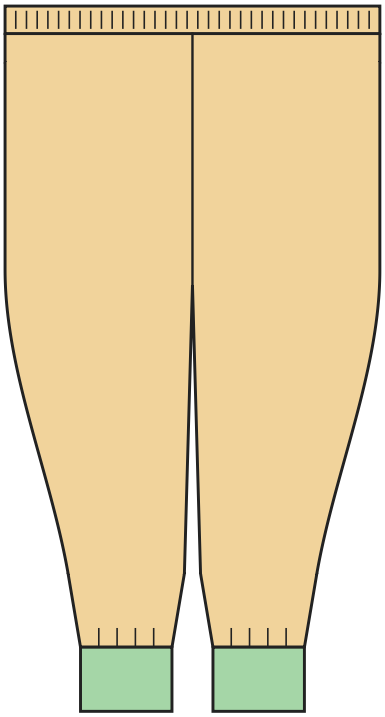
1. Press all seams with steam
2. Do not press down hard on sweat.



Done!



Front view



Back view

After the test garment: Try the test garment on the child: length, width at the hip, the crotch and the waistband. Adjust the measurements or options and print the pattern again before you cut the good fabric.

Harem pants · size 122

harem (wide, dropped crotch) fit, Sweat / French Terry (25 % stretch) · original parametric draft. The pattern is yours to use.

> **Important:** With a new pattern, first sew a **test garment** from inexpensive fabric with similar stretch and check the fit.

Files

- PDF A4 tiles to glue (12 pages + overview): `harem_pants_size122_a4_tiled.pdf`
- Preview A4 tiles: `harem_pants_size122_a4_preview.png`
- SVG master 1:1: `harem_pants_size122_master_1to1.svg`
- Cutting layout: `cutting_layout.png`
- Sewing instructions (PDF): `sewing_instructions.pdf`
- Sewing instructions (Markdown): `sewing_instructions.md`

Printing

- Always print at **100 % / "Actual size"** (not "Fit to page").
- Measure the **5 × 5 cm** control square before you cut.
- Plotter (610 mm roll): the page is 600 mm wide in portrait. In the print dialog choose a custom paper size with exactly this width and length, and turn rotation/scaling off.

Body measurements

Body measurement	Value	Source
Body height	122.0 cm	table
Chest	62.0 cm	table
Waist	56.0 cm	table
Hip	66.0 cm	table
Head	52.0 cm	table
Neck	28.5 cm	table
Inseam	54.0 cm	table
Side length	72.0 cm	table
Arm length	47.0 cm	table
Back length	29.5 cm	table
Shoulder width	10.0 cm	table
Thigh	37.0 cm	table
Ankle	17.8 cm	table

Finished measurements (virtually sewn)

Finished measurement	Value
Hip width	85.8 cm
Thigh width	55.5 cm
Leg width at mid-leg	42.0 cm
Leg opening	20.0 cm
Waistband relaxed	50.4 cm
Waistband stretched (pants top edge)	83.0 cm
Outseam	75.5 cm
Inseam	48.2 cm
Front rise	27.3 cm
Back rise	29.1 cm
Waistband height	3.0 cm
Cuff height	7.0 cm

Finished measurement	Value
Crotch seam total	64.2 cm

Pattern pieces

Piece	Cutting	Fabric	Seam allowance
Front leg	cut 2 (mirrored pair)	Main fabric	seam allowance 1.0 cm included
Back leg	cut 2 (mirrored pair)	Main fabric	seam allowance 1.0 cm included
Waistband (elastic)	cut 1 on fold	Main fabric	seam allowance 1.0 cm included
Leg cuff	cut 2	Rib knit	seam allowance 1.0 cm included

Fabric needed

- **Main fabric:** about 0.90 m at 150 cm width (folded selvage to selvage, cut on fold pieces at the fold, incl. 10 % for shrinkage; single layer would need 0.90 m)
- **Rib knit:** about 0.20 m at 70 cm width (single layer, incl. 10 % for shrinkage)
- Elastic 20 mm wide, length 50 cm (+2 cm overlap for joining)

Cutting layout: see `cutting_layout.png` (grainline arrow parallel to the selvage, "flipped" = place the piece face down, "×2" = cut through both layers of the folded fabric, which gives a mirrored pair, "(unfolded)" = cut-on-fold piece shown opened out).

Cut on fold (Waistband (elastic)): the paper piece is only one half. Its fold edge (dash-dot line, bracket arrows "PLACE ON FOLD") is the centre of the piece: it gets no seam allowance and is never cut. On the folded fabric, lay the fold edge exactly on the fabric fold, pin it and cut through both layers; opened out you get the whole piece. Where the cutting layout shows a piece "(unfolded)" on a single layer, fold the fabric along the grain just wide enough for the piece and place the fold edge on that fold, or trace the paper piece, flip it over along the fold edge and trace the second half.

Validation report

Sewability check

Area	Check	Target	Actual	Δ	Result	Note
Seam pairs	Side seam front ↔ back	66.0 cm	66.0 cm	+0.0 mm	✓ OK	
Seam pairs	Inseam front ↔ back	41.7 cm	41.7 cm	+0.0 mm	✓ OK	
Seam pairs	Front crotch seam (left ↔ right piece)	26.9 cm	26.9 cm	+0.0 mm	✓ OK	
Seam pairs	Back crotch seam (left ↔ right piece)	31.3 cm	31.3 cm	+0.0 mm	✓ OK	
Seam pairs	Waistband ↔ pants top edge	83.0 cm	83.0 cm	+0.0 mm	✓ OK	
Seam pairs	Leg cuff ↔ leg opening (stretched)	70–85 % of 28.5 cm	20.0 cm = 70 %	-85.5 mm	✓ OK	rib band is stretched while sewing
Angles/transitions	Centre front ↔ waistband	90°	90.0°	+0.0°	✓ OK	
Angles/transitions	Centre back ↔ waistband	90°	90.0°	-0.0°	✓ OK	
Angles/transitions	Crotch ↔ inseam (front + back)	180°	180.9°	+0.9°	✓ OK	
Angles/transitions	Inseam ↔ hem front	90°	90.1°	+0.1°	✓ OK	
Angles/transitions	Hem ↔ side seam front	90°	90.1°	+0.1°	✓ OK	
Angles/transitions	Inseam ↔ hem back	90°	90.2°	+0.2°	✓ OK	
Angles/transitions	Hem ↔ side seam back	90°	90.2°	+0.2°	✓ OK	

Area	Check	Target	Actual	Δ	Result	Note
Angles/transitions	Side seam ↔ waistband back	90°	90.1°	+0.1°	✓ OK	
Angles/transitions	Side seam ↔ waistband (front + back)	180°	180.1°	+0.1°	✓ OK	
Common practice	Hip ease for fit 'harem'	25 – 45 %	30 %	+0 %	✓ OK	guide value for this fit
Measurements	Hip circumference pattern = body + ease	85.8 cm	85.8 cm	+0.0 %	✓ OK	body 66.0 cm + +30 % ease
Measurements	Top edge fits over the hip	≥ 67.3 cm	95.4 cm	+281.1 mm	✓ OK	top edge incl. usable stretch ≥ hip
Measurements	Front rise plausible	14.6 cm – 31.4 cm	27.3 cm	+0.0 mm	✓ OK	body rise 18.0 cm, harem drop 7.3 cm
Measurements	Inseam length plausible	39.0 cm – 60.5 cm	55.5 cm	+0.0 mm	✓ OK	body inseam 54.0 cm
Measurements	Foot fits through the leg opening	≥ 24.1 cm	29.5 cm	+54.8 mm	✓ OK	heel-instep circumference ≈ 1.35 × ankle
Geometry	Front leg: outline & seam allowance	closed, SA present	ok		✓ OK	seam allowance 1.0 cm included
Labels	Front leg: grainline & cutting	present	cut 2 (mirrored pair)		✓ OK	Main fabric
Geometry	Back leg: outline & seam allowance	closed, SA present	ok		✓ OK	seam allowance 1.0 cm included
Labels	Back leg: grainline & cutting	present	cut 2 (mirrored pair)		✓ OK	Main fabric
Geometry	Waistband (elastic): outline & seam allowance	closed, SA present	ok		✓ OK	seam allowance 1.0 cm included
Labels	Waistband (elastic): grainline & cutting	present	cut 1 on fold		✓ OK	Main fabric
Geometry	Leg cuff: outline & seam allowance	closed, SA present	ok		✓ OK	seam allowance 1.0 cm included
Labels	Leg cuff: grainline & cutting	present	cut 2		✓ OK	Rib knit
Geometry	Waistband (elastic): cut-on-fold edge	straight, no SA, 90° corners, on grain	ok, corners ≤ 0.0° off 90°		✓ OK	Centre front
Labels	Front leg: notches	on the cutting line	3 notch mark(s)		✓ OK	
Labels	Back leg: notches	on the cutting line	6 notch mark(s)		✓ OK	

Result: 32 OK, 0 warning(s), 0 error(s): the pattern is sewable.

Construction values

Override any of these with the option `construction`, values in mm (e.g. `{"construction": {"hip_width": 858}}`).

Name	Value	Derived from	Note
`hip_width`	85.8 cm	hip × (1 + 30 % ease)	
`front_back_offset`	0.5 cm	front quarter narrower, back quarter wider by this amount	
`front_width`	20.9 cm	hip_width / 4 − front_back_offset	

Name	Value	Derived from	Note
back_width	21.9 cm	hip_width / 4 + front_back_offset	
front_crotch_extension	4.3 cm	5 % of hip_width	
back_crotch_extension	8.6 cm	10 % of hip_width	
rise	24.3 cm	(side length − inseam) + seat ease + harem drop − waistband height	
inseam_finished	48.2 cm	inseam − allowance to the floor − harem drop + blousing	
hem_width	28.5 cm	cuff length / cuff stretch ratio	

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